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Scope-Bound Cybersecurity Evidence Passports With Independent Review and Living Assurance Lineage

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ABSTRACT Organizations routinely exchange cybersecurity reports, attestations, scan results, and provenance records. Once separated from the processes that produced them, however, these records may retain a valid digest while losing the context needed to judge scope, exclusions, reviewer independence, currency, and supersession. This study develops Sampling Sigma, a design science prototype for scope-bound cybersecurity evidence passports. Confidential evidence remains off-chain; a portable passport binds the artifact digest to its subject, declared scope, limitations, validity period, issuer, independent review, conflict disclosure, lifecycle status, and lineage. The prototype comprises a Solidity contract, a local verifier, and a reproducible evaluation harness. All 15 contract tests passed. In controlled experiments, the verifier detected every one of 5,000 seeded single-bit mutations, and the reference lifecycle model rejected all 21 forbidden transitions among 25 ordered state pairs. These results demonstrate internal consistency under the reported conditions. They do not establish semantic truth, security certification, interoperability, usability, or regulatory compliance. The study therefore separates integrity, provenance, validation, and economics and specifies a preregisterable comparison of raw digest exchange, signed centralized records, credential-only passports, and an optional shared registry.

INDEX TERMS Blockchain, cybersecurity assurance, design science, evidence provenance, independent review, living assurance, privacy, smart contracts, verifiable credentials.

I. INTRODUCTION

Cybersecurity assurance depends on more than preserving a file's bytes. A relying party must also know what was assessed, what remained outside the assessment, who approved the evidence, whether the reviewer was independent, and whether a later record superseded the one presented. Secure timestamping [1], supply-chain attestation [2], verifiable credentials [3], and cybersecurity-control assessment [4] address portions of this problem. Used separately, however, they do not provide a compact decision record that carries integrity, scope, limitations, review, validity, and lineage together.

Sampling Sigma examines whether a portable evidence passport can preserve that decision context without transferring the underlying confidential artifact. The design deliberately treats a shared ledger as optional. Its inclusion must be supported by evidence that cross-organizational coordination improves beyond what a signed centralized record can provide.

The work makes four contributions. First, it defines a minimum passport schema that records both assurance claims and their boundaries. Second, it separates the issuer, reviewer organization, relying party, and economic roles. Third, it models rejection, expiry, revocation, and supersession as explicit lifecycle events. Finally, it couples a reproducible prototype evaluation with a preregisterable protocol for subsequent human-subject testing.

II. RESEARCH QUESTIONS AND RELATED WORK

A. RESEARCH QUESTIONS

- RQ1: Which fields prevent a fingerprint from being mistaken for proof of truth or full-scope assurance?

- RQ2: Can organizational separation and conflict commitments make silent self-approval detectable?
- RQ3: Can explicit validity and lineage reduce acceptance of stale or superseded evidence?
- RQ4: Can local verification preserve confidentiality while remaining usable?
- RQ5: Under which multi-organization conditions does a shared registry add value over signed centralized records?

B. RESEARCH GAP

Existing standards supply many of the required building blocks. Haber and Stornetta established tamper-evident document timestamping [1]. in-toto provides authenticated software-supply-chain provenance [2], and SLSA defines progressive artifact-integrity practices [5]. The W3C Verifiable Credentials Data Model contributes portable claims, issuer proofs, and status semantics [3]. NIST SP 800-53A frames evidence-driven control assessment [4], while C2PA offers a related model of provenance and trust [6]. What remains underdeveloped is a compact record that assembles these capabilities into enough decision context for a relying party to judge a cybersecurity assurance claim without receiving the confidential source artifact.

III. DESIGN SCIENCE METHOD

The study follows the design science cycle described by Hevner et al. [7], progressing from problem identification and requirements derivation through construction, demonstration, evaluation, and communication. Requirements were synthesized from guidance on assessment, secure development, provenance, credentials, and threat modeling [3]-[6], [8]. Evaluation proceeds in three stages: deterministic contract tests, controlled mutation and reference state-machine experiments, and a planned comparative human study. Measured results are currently available for the first two stages only. Table I summarizes the resulting requirements and protected failure modes.

TABLE I
DESIGN REQUIREMENTS AND PROTECTED FAILURE MODES

ID	Requirement	Failure
R1	Bind artifact, subject, scope, limitations, and period	Context collapse
R2	Separate issuer and reviewer organization	Self-approval
R3	Commit conflicts and review evidence	Hidden dependency
R4	Expose validity, rejection, revocation, supersession	Stale assurance
R5	Keep evidence off-chain; verify locally	Disclosure
R6	Separate validity from token economics	Economic capture

IV. ARCHITECTURE AND DATA MODEL

A. ASSURANCE PLANES

The architecture distributes responsibility across five assurance planes and an optional economic plane. Source artifacts remain in an encrypted evidence vault. The passport service canonicalizes, hashes, validates, and signs the portable record, while an independence graph represents organizations and disclosed relationships. An assurance registry stores minimum commitments and lifecycle events. At the point of use, the local verifier recalculates the digest, validates the signature and status, checks validity and supersession, and applies the relying party's policy. The optional economic layer is structurally excluded from decisions about evidentiary truth. Table II groups the fields carried by the evidence passport.

TABLE II
EVIDENCE PASSPORT FIELD GROUPS

Group	Required fields
Identity	Schema version, passport ID, subject commitment
Integrity	Algorithm, artifact digest, canonicalization profile
Context	Domain, assessment type, scope and limitation commitments
Review	Issuer, reviewer organization, conflict and evidence commitments
Lifecycle	Issued time, valid-until, status, revocation, supersession
Proof	Signature suite, key identifier, policy and status endpoints

A successful digest check establishes byte equality with the committed artifact. Correctness, completeness, authorization, reviewer competence, and continuing security require separate evidence. The verifier states this boundary during the decision process so that a valid digest cannot be presented as broader proof.

B. FORMAL COMMITMENTS AND DECISION PREDICATE

Let x be a JSON evidence passport excluding its proof. The portable commitment uses the JSON Canonicalization Scheme [12] followed by SHA-256 as specified by FIPS 180-4 [11].

$$C(x) = SHA\text{-}256(JCS(x)) \quad (1)$$

For passport P, artifact A, and decision time t, acceptance is conjunctive rather than score-based. Each indicator equals one only when its named check passes.

$$V = H \wedge S \wedge C \wedge R \wedge T \wedge Q \quad (2)$$

H recomputes the artifact commitment; S validates the issuer proof; C requires the relying party to acknowledge scope and limitations; R checks reviewer identity, organizational separation, and conflict disclosure; T rejects expired, revoked, rejected, or superseded passports; and Q applies local policy. A zero in any term produces rejection. This model prevents confidence scores from overriding a failed mandatory control. Table III summarizes the threats used to challenge these checks.

V. THREAT MODEL

TABLE III
THREATS, MITIGATIONS, AND RESIDUAL RISKS

Threat	Mitigation	Residual risk
Substitution	Local digest and signature check	Compromised endpoint
Scope inflation	Mandatory scope and limitations	Deceptive wording
Self-review	Role and organization separation	Collusion
Stale evidence	Expiry, revocation, supersession	Publication delay
Data leakage	Off-chain evidence and minimization	Metadata correlation
Admin compromise	Role separation, multisignature target, audit	Key compromise

VI. PROTOTYPE IMPLEMENTATION

A. SMART CONTRACT

The prototype uses Solidity 0.8.28 and OpenZeppelin components [9]. SamplingSigmaAssuranceCore records subject, scope, limitations, validity, reviewer organization, conflict commitment, rejection, revocation, and supersession. A browser-local verifier hashes files and produces a JSON passport without uploading the source. The separate token candidate is not required for evidence verification, and no public sale, exchange listing, or production deployment is authorized by this research artifact.

B. VERIFICATION ALGORITHM

ALGORITHM 1
LOCAL PASSPORT VERIFICATION

```
INPUT: passport P, local artifact A, time t, policy Q
require schema(P) and canonicalizationProfile(P)
require SHA256(A) = P.artifactHash
require verifySignature(P.proof, issuerKey(P))
require scopeAccepted(P.scope, P.limitations)
require independentReviewer(P) and conflictChecked(P)
require P.status = VALIDATED and t <= P.validUntil
require notRevoked(P) and notSuperseded(P)
require Q(P) = true; return ACCEPT
on any failed requirement: return REJECT with reason
```

The verifier applies fail-closed semantics. Malformed records, unknown algorithms, missing commitments, failed proofs, unavailable status information, and unmet policy conditions each produce a rejection with a stated reason. The implementation uses explicit custom errors and emits lifecycle events to preserve an auditable decision trail.

C. REPRODUCIBILITY PACKAGE

The reproducibility package records the source revision, dependency lock file, test output, evaluation script, raw JSON results, machine environment, and document checksum. The public research artifact is available at https://github.com/SriMurukutla/sampling-sigma-evidence-passports. The evaluated submission snapshot is anchored by commit 3df129504776c6313f0b4a03ec0f242caa88a94d; a separate archival DOI may be added after depositing the tagged release in a long-term repository.

VII. MEASURED PROTOTYPE RESULTS

A. EXPERIMENTAL SETUP

Study SS-DSR-PROTOTYPE-2026-10-29 executed three deterministic experiments and the repository contract suite. The mutation experiment generated 5,000 seeded byte strings, flipped one indexed bit in each string, and compared SHA-256 digests. The lifecycle reference model enumerated all 25 ordered pairs over Submitted, Validated, Rejected, Revoked, and Superseded. The performance test hashed a deterministic one-mebibyte payload 256 times using a monotonic nanosecond timer. Contract tests ran through the repository's npm test command. Tables IV and V report the observed measurements and contract-suite composition.

B. OBSERVED RESULTS

TABLE IV
OBSERVED PROTOTYPE MEASUREMENTS

Experiment	Observed result	Boundary
Contract suite	15 passing; exit 0	Not an audit
Mutation corpus	5000/5000 detected	Byte changes only
Lifecycle model	25 pairs; 21 forbidden rejected	Reference model
SHA-256 benchmark	Median 2.1396 ms; p95 2.7037 ms	Single machine/run

The controlled corpus detected 100% of 5,000 injected single-bit mutations. Under a simplifying independent-Bernoulli model, the exact two-sided 95% lower confidence bound for detection in this generator is 99.9263%.

$$L95 = 0.025^{\wedge}(1/5000) = 0.999263 \quad (3)$$

The bound describes only this seeded mutation generator; it does not estimate SHA-256 collision resistance, validate semantic truth, or establish an operational false-acceptance probability. The benchmark used Python 3.12.14 on macOS-15.4.1-x86_64-i386-64bit and measured 449.01 MiB/s for 256 one-mebibyte payloads. The latency distribution is reported by median and p95 because no normality assumption was made.

TABLE V
CONTRACT SUITE COMPOSITION

Component	Tests	Primary behaviors
Assurance core	4	Commitments, independent review, supersession
Research core	6	Evidence validation, authorization, immutability
Token candidate	5	Fixed allocation, role separation, pause controls

All 15 repository tests passed in the recorded run. Four tests exercised the assurance core, six addressed research-record governance, and five covered the separate token candidate. The token results are included to document repository completeness; they do not answer the assurance research

questions. The prototype has not yet undergone an independent audit, user study, interoperability assessment, production-availability study, or legal evaluation.

VIII. COMPARATIVE EVALUATION PROTOCOL

A controlled study will compare four conditions: raw digest exchange; a signed centralized JSON record; a W3C Verifiable Credential passport without a ledger; and Sampling Sigma with an optional shared registry. Participants will issue and verify identical evidence packages containing altered, expired, superseded, conflicted, and out-of-scope cases. An exploratory pilot of 24 to 30 participants will test the instruments and estimate variance. A confirmatory sample will then be determined by preregistered power analysis using a declared smallest effect of interest. Table VI defines the planned outcomes and analyses.

TABLE VI
PLANNED PRIMARY OUTCOMES AND ANALYSIS

Outcome	Definition	Analysis
Accuracy	Correct accept/reject decision	Mixed logistic model
Time	Seconds to defensible decision	Survival/log-normal model
Critical error	Invalid accepted or valid rejected	Risk difference and CI
Completeness	Required context interpreted	Rubric and agreement
Privacy	Evidence bytes transmitted	Network capture
Usability	SUS and task workload	CI and effect size

The confirmatory protocol will fix primary outcomes, exclusions, and analysis before data collection. Holm correction will control multiplicity. Effect sizes and 95% confidence intervals will accompany p-values. Missing data, deviations, adverse events, and exclusions will be reported. Robust or nonparametric sensitivity analyses will address distributional violations.

IX. PROSPECTIVE RESULTS AND DECISION RULES

The following decision rules concern prospective outcomes and should not be read as observed findings. A shared registry will be retained only if the comparative study identifies a measurable advantage over simpler designs.

- If the lower 95% confidence limit for decision-accuracy improvement is above zero without increased critical errors, proceed to a larger pilot.
- If the credential-only passport is within the preregistered equivalence margin, prefer that simpler architecture.
- If users over-trust successful digest checks and the critical misconception rate exceeds 5%, redesign the verifier language and workflow.
- If reviewer independence cannot be verified, do not claim independent-review assurance.
- If metadata-correlation risk exceeds the approved privacy threshold, remove or reduce public metadata.

X. VALIDITY, ETHICS, AND REPRODUCIBILITY

A. VALIDITY

Construct validity is protected by distinguishing digest equality from truth. Internal validity uses identical task packages, randomized order, logged deviations, and blinded scoring where practical. External validity is limited: results from a controlled pilot will not be generalized to all sectors or jurisdictions.

B. ETHICS AND PRIVACY

No real customer secrets or personal data will enter a public registry. Human-participant research requires the applicable institutional determination, informed consent where required, minimization, retention limits, and a withdrawal process. Security findings will follow coordinated disclosure.

C. REPRODUCIBILITY

The release will include schemas, fixtures, source code, dependency versions, raw measurements, test output, analysis scripts, and replication instructions where legally permitted. IEEE's Code Ocean and DataPort integrations provide possible archival channels [10].

XI. GOVERNANCE AND ECONOMIC SEPARATION

The assurance protocol can operate without a tradable token. Any token distribution, sale, custody, listing, compensation, staking, or investment-oriented communication requires separate legal, tax, sanctions, consumer-protection, and securities analysis. Evidence validity must never depend on token ownership, price, or voting power. The article evaluates assurance architecture and does not make an investment claim.

XII. LIMITATIONS

- The measurements are local prototype results and have not been independently replicated.
- The lifecycle experiment checks a reference model and is not a formal contract-equivalence proof.
- Identity, reviewer competence, organizational independence, and conflicts require external validation.
- No field evidence yet establishes decision improvement, usability, interoperability, or economic value.
- Public registries can introduce permanence, cost, governance, and metadata-correlation risks.

XIII. CONCLUSION

Sampling Sigma provides a structured way to carry artifact integrity, decision context, independent review, and lifecycle state across organizational boundaries while keeping the source evidence under local control. The prototype behaved consistently in the reported contract, mutation, and state-transition tests. That result is encouraging but narrow. Claims about practical usefulness must await the planned comparison with simpler signed records and credential-only passports. Continued use of the shared-registry design should depend on

demonstrated gains in decision quality, privacy, continuity, or cross-organizational coordination.

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DATA AVAILABILITY.

The prototype source, dependency lock file, complete contract tests, evaluation script, raw JSON measurements, test transcript, checksum manifest, and submission-candidate manuscript are publicly available at https://github.com/SriMurukutla/sampling-sigma-evidence-passports. The paper's evaluated snapshot is commit 3df129504776c6313f0b4a03ec0f242caa88a94d. The repository is public under the MIT License for code; no independent audit, production approval, or regulatory determination is implied. Table VII lists the minimum reproducibility artifacts.

APPENDIX A. REPRODUCIBILITY ARTIFACTS

TABLE VII
REPRODUCIBILITY ARTIFACTS

Artifact	Purpose
contracts/SamplingSigmaAssuranceCore.sol	Assurance record and lifecycle implementation
test/SamplingSigmaAssuranceCore.ts	Assurance-core behavioral tests
tools/research_evaluation.py	Mutation, lifecycle, benchmark, and test harness
evaluation/actual-prototype-results.json	Machine-readable observed measurements
evaluation/hardhat-test-output.txt	Recorded contract-test output

Minimum replication sequence: install the locked dependencies; run npm test; run python tools/research_evaluation.py; compare the resulting JSON study identifier, environment, test exit code, case counts, and timing statistics with the reported artifact. Timing values are expected to vary by machine; logical counts and rejection outcomes should not.

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